

Bilateral Sharpness-Aware Minimization for Flatter Minima

Jiaxin Deng¹, Junbiao Pang¹, Baochang Zhang², and Qingming Huang³, *Fellow, IEEE*

Abstract—Sharpness-aware minimization (SAM) enhances generalization by minimizing max-sharpness (MaxS). Despite its practical success, we empirically found that the MaxS behind SAM’s generalization enhancements faces the “flatness indicator problem” (FIP), where SAM only considers the flatness in the direction of gradient ascent. This leads to high Hessian eigenvalues for the deep neural network (DNN), indicating insufficient flatness in the solution region. A better flatness indicator (FI) would lower these Hessian eigenvalues, resulting in a flatter minimum and improved generalization of the network. Because SAM is inherently a greedy search method. In this article, we propose to utilize the difference between the training loss and the minimum loss over the neighborhood surrounding the current weight, which we denote as min-sharpness (MinS). By merging MaxS and MinS, we create a better FI that indicates a flatter direction during the optimization. Specifically, we combine this FI with SAM into the proposed bilateral SAM (BSAM), which finds a flatter minimum than SAM. The theoretical analysis demonstrates that BSAM converges to a local minimum. Extensive experiments demonstrate that BSAM offers superior generalization performance and robustness compared to vanilla SAM across various tasks, i.e., classification, transfer learning, human pose estimation, semantic segmentation, and network quantization.

Index Terms—Flatness, flatter minima, generalization, sharpness-aware minimization (SAM).

I. INTRODUCTION

DEEP neural networks (DNNs) have achieved impressive results across various fields but are often over-parameterized, leading to overfitting and poor generalization when trained with standard loss functions [1]. Recent research has shown that the loss landscape is complex and nonconvex with many local minima of different generalization abilities

[2]. Consequently, many studies have investigated the relationship between the geometry of the loss landscape and the generalization of a model [3], [4]. Optimizers typically need to converge to a flat minimum [5], [6] for good generalization performance. However, the complex loss landscape of over-parameterized DNNs, with many sharp minima, makes it difficult for optimizers to find a flat minimum. For example, stochastic gradient descent (SGD) and some of its variants serve as an implicit regularization that favors flat minima [7]. However, researchers prefer to use optimizers in an explicit manner to find the smooth loss landscapes and a flat minimum.

Sharpness-aware minimization (SAM) [1] explicitly promotes a flat minimum by jointly minimizing the loss value and loss sharpness, leading to improved generalization performance. SAM and its variants have demonstrated state-of-the-art (SOTA) performance across various applications, such as classification [8], transfer learning [9], [10] and domain generalization [11].

However, we found that SAM faces the “flatness indicator problem” (FIP), which results in the minimum region where the solution lies not being sufficiently flat. Specifically, SAM’s flatness indicator (FI) considers only the sharpness in the gradient ascent direction, while ignoring the flatness along the gradient descent direction. As a result, SAM tends to converge to a locally flat minimum that is sharp on one side during training. Because SAM is inherently a greedy search method that selects the flattest direction at each update step, designing an effective FI to measure the flatness of the neighborhood around the current parameter point is critical. A more accurate FI can better guide the optimizer toward a flatter minimum. That is, a better FI would lead to better generalization of neural networks.

In this article, we propose a more effective FI to guide the search for flatter minima than SAM. We introduce min-sharpness (MinS), which quantifies sharpness along the gradient descent direction. Specifically, MinS is defined as the difference between the current training loss and the minimum loss within a local neighborhood in parameter space. By combining max-sharpness (MaxS) and MinS, we formulate a novel FI, bilateral sharpness (BiS), and develop a new optimization method called bilateral SAM (BSAM). BSAM is capable of identifying flatter directions early in training and ultimately converges to a flatter minimum, leading to improved generalization performance. To address potential conflicts between the MinS gradient and the descent direction of the loss, we introduce a radius decay strategy for MinS. Our theoretical analysis guarantees that BSAM converges to a local minimum.

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Jiaxin Deng and Junbiao Pang are with the Faculty of Information Technology, Beijing University of Technology, Beijing 100124, China (e-mail: junbiao_pang@bjut.edu.cn).

Baochang Zhang is with the School of Artificial Intelligence, Beihang University, Beijing 100191, China (e-mail: bczhang@buaa.edu.cn).

Qingming Huang is with the University of Chinese Academy of Sciences, Chinese Academy of Sciences (CAS), Beijing 100049, China, and also with the Institute of Computing Technology, CAS, Beijing 100190, China (e-mail: qmhuang@ucas.ac.cn).

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In summary, our contributions are as follows.

- 1) We found that BiLS is a better FI than MaxS. Based on this finding, we propose optimizing the BiLS of the loss landscape within the local parameter space. Our results highlight the crucial role of BiLS in achieving flatter minima for vision tasks.
- 2) We propose the BSAM method, which simultaneously optimizes the MaxS, MinS, and the current training loss to find a flatter minimum. Experimental analysis of the top eigenvalues of the Hessian reveals that BSAM achieves flatter minima compared to SAM. By applying BSAM to various tasks such as classification, transfer learning, human pose estimation, and semantic segmentation, BSAM significantly enhances model generalization performance.

II. RELATED WORKS

A. Background of SAM

Foret et al. [1] introduced the SAM method to enhance model generalization. The optimization objective of SAM is as follows:

$$\min_{\mathbf{w}} [(L^{\max}(\mathbf{w}) - L(\mathbf{w})) + L(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2] \\ L^{\max}(\mathbf{w}) = \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} L(\mathbf{w} + \boldsymbol{\varepsilon}) \quad (1)$$

where $\mathbf{w}$ represents the weights of the network, $\boldsymbol{\varepsilon}$ represents the perturbation of weights $\mathbf{w}$ in a Euclidean ball with the radius ρ ($\rho > 0$), $L(\cdot)$ is the loss function, and $\lambda \|\mathbf{w}\|_2^2$ is a standard L2 regularization term. The sharpness in SAM, referred to as MaxS, is defined as follows:

$$h^{\max}(\mathbf{w}) = \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} L(\mathbf{w} + \boldsymbol{\varepsilon}) - L(\mathbf{w}) \quad (2)$$

where $h^{\max}(\mathbf{w})$ quantifies the sharpness of $L(\mathbf{w})$ by measuring how quickly the training loss can be decreased by moving from $\mathbf{w}$ to nearby weight values.

In order to obtain $h^{\max}(\mathbf{w})$, SAM utilizes Taylor expansion to search for the maximum perturbed loss ($L^{\max}(\mathbf{w})$) in local parameter space as follows:

$$\arg \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} L(\mathbf{w} + \boldsymbol{\varepsilon}) \approx \arg \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} L(\mathbf{w}) + \boldsymbol{\varepsilon}^T \nabla_{\mathbf{w}} L(\mathbf{w}) \\ = \arg \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} \boldsymbol{\varepsilon}^T \nabla_{\mathbf{w}} L(\mathbf{w}). \quad (3)$$

By solving (3), SAM obtains the perturbation as follows:

$$\hat{\boldsymbol{\varepsilon}} = \rho \nabla_{\mathbf{w}} L(\mathbf{w}) / \|\nabla_{\mathbf{w}} L(\mathbf{w})\|. \quad (4)$$

This perturbation follows the gradient ascent direction and maximizes the perturbed loss ($L^{\max}(\mathbf{w})$). Substituting the perturbation $\hat{\boldsymbol{\varepsilon}}$ back into (1) and differentiating, we have

$$\nabla_{\mathbf{w}} L^{\max}(\mathbf{w}) \approx \nabla_{\mathbf{w}} L(\mathbf{w} + \hat{\boldsymbol{\varepsilon}}(\mathbf{w})) \\ = \frac{d(\mathbf{w} + \hat{\boldsymbol{\varepsilon}}(\mathbf{w}))}{d\mathbf{w}} \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\boldsymbol{\varepsilon}}(\mathbf{w})} \\ = \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\boldsymbol{\varepsilon}}(\mathbf{w})} + \frac{d\hat{\boldsymbol{\varepsilon}}(\mathbf{w})}{d\mathbf{w}} \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\boldsymbol{\varepsilon}}(\mathbf{w})}. \quad (5)$$

By dropping the second-order terms in (5), SAM calculates the gradient at $\mathbf{w} + \hat{\boldsymbol{\varepsilon}}$ as follows:

$$\nabla_{\mathbf{w}} L^{\max}(\mathbf{w}) \approx \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\boldsymbol{\varepsilon}}}. \quad (6)$$

Finally, SAM uses the gradients from (6) for optimization.

B. Variants of SAM

SAM and its variants can be roughly divided into two categories.

The first category focuses on enhancing the model's generalization ability. To address the scale-dependency problem of sharpness, where sharpness measures are sensitive to model parameter rescaling, thereby weakening the correlation between sharpness and the generalization gap. Kwon et al. proposed adaptive SAM (ASAM) [8], which introduces adaptive sharpness to mitigate this issue and improve generalization. Zhuang et al. [10] pointed out that SAM did not consistently favor the flat minima. They then proposed surrogate gap guided SAM (GSAM) to improve the generalization ability by minimizing the perturbed loss and the proposed surrogate gap, which measures the difference between the maximum loss within the neighborhood and the loss at the center point. Zhao et al. [12] proposed a variant of SAM by penalizing the gradient norm based on the observation that a sharper valley of the loss landscape tends to have the gradient with a larger norm. Zhang et al. [13] introduced the first-order flatness, which measures the maximal gradient norm within a perturbation radius of the center point. Based on this, they proposed gradient norm aware minimization (GAM), a new training approach that seeks a flatter loss landscape with uniformly small curvature, leading to improved generalization. Li et al. [14] introduced friendly-SAM (F-SAM), which refines SAM by removing the detrimental effects of the full gradient component of the whole dataset. In summary, these methods improve SAM from different perspectives, such as the definition of sharpness and the optimization process of sharpness, enhancing the generalization performance of the trained models. However, they both considered only unidirectional flatness.

The second category aims to enhance the optimization efficiency of SAM. The optimization efficiency of SAM has limited its large-scale applications since SAM needs two forward and two backward passes per optimization step. To address this, several methods have been proposed to accelerate SAM. For instance, Zhao et al. [15] proposed randomized SAM (G-RST), which reduces the computational burden by randomly choosing between base optimization algorithms and sharpness-aware methods at each iteration. This approach lowers the total number of forward-backward propagation required. Liu et al. [16] proposed LookSAM, which accelerates SAM by reducing the frequency of perturbation computations. Jiang et al. [17] introduced an adaptive policy to employ SAM (AE-SAM) to speed up SAM by adaptively choosing when SAM is used based on the loss landscape. Ni et al. [18] proposed K-SAM, which reduces SAM's computational cost by computing gradients only with the top- k samples with the highest loss. Du et al. [9] proposed efficient SAM (ESAM) to accelerate SAM by two strategies: stochastic weight perturbation, which approximates sharpness by perturbing a subset of weights, and sharpness-sensitive data selection, which uses a carefully chosen data subset to optimize the loss. Becker et al. [19] proposed momentum-SAM (MSAM) without requiring additional forward and backward computations by using the momentum direction as an approximation of the perturbation direction. In summary, these methods greatly improve the computational efficiency of SAM-based methods, but most

of them come at the cost of sacrificing model generalization performance.

III. METHOD

A. Min-Sharpness

MaxS in (2) just captures the difference between the maximum loss of perturbed points and the loss at the current parameter point, promoting flatness from a single direction.

To describe flatness along the gradient descent direction, we define MinS as follows:

$$h^{\min}(\mathbf{w}) = L(\mathbf{w}) - \min_{\|\boldsymbol{\varepsilon}^{\min}\| \leq \rho^{\min}} L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min}) \quad (7)$$

where $\boldsymbol{\varepsilon}^{\min}$ represents weight perturbations in Euclidean ball with radius $\rho^{\min}$. MinS measures how quickly the training loss decreases as the parameters move from $\mathbf{w}$ to a nearby value.

We obtain the optimal $\boldsymbol{\varepsilon}^{\min}$ that minimizes the loss $L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min})$ in (7) via a first-order Taylor expansion of $L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min})$ around 0 as follows:

$$\begin{aligned} \hat{\boldsymbol{\varepsilon}}^{\min} &= \arg \min_{\|\boldsymbol{\varepsilon}^{\min}\| \leq \rho^{\min}} L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min}) \\ &\approx \arg \min_{\|\boldsymbol{\varepsilon}^{\min}\| \leq \rho^{\min}} L(\mathbf{w}) + \boldsymbol{\varepsilon}^{\min \top} \nabla_{\mathbf{w}} L(\mathbf{w}) \\ &\approx \arg \min_{\|\boldsymbol{\varepsilon}^{\min}\| \leq \rho^{\min}} \boldsymbol{\varepsilon}^{\min \top} \nabla_{\mathbf{w}} L(\mathbf{w}). \end{aligned} \quad (8)$$

$L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min})$ reaches its minimum value when

$$\hat{\boldsymbol{\varepsilon}}^{\min} = -\rho^{\min} \frac{\nabla_{\mathbf{w}} L(\mathbf{w})}{\|\nabla_{\mathbf{w}} L(\mathbf{w})\|} \quad (9)$$

where $\nabla_{\mathbf{w}} L(\mathbf{w})$ is the gradient at $\mathbf{w}$, and $\|\cdot\|$ represents the L2-norm. We observe that the direction of $\hat{\boldsymbol{\varepsilon}}^{\min}$ is along the gradient descent direction, which is opposite to the perturbation direction in vanilla SAM.

B. Bilateral SAM

We propose a method to jointly optimize the training loss ($L(\mathbf{w})$), MaxS [see (2)], and MinS [see (7)] to find flatter minima. The overall loss function can be written as follows:

$$\begin{aligned} \hat{L}(\mathbf{w}) &= \underbrace{L(\mathbf{w})}_{\text{Training loss}} \\ &+ \underbrace{\left[\max_{\|\boldsymbol{\varepsilon}^{\max}\| \leq \rho^{\max}} L(\mathbf{w} + \boldsymbol{\varepsilon}^{\max}) - \min_{\|\boldsymbol{\varepsilon}^{\min}\| \leq \rho^{\min}} L(\mathbf{w} + \boldsymbol{\varepsilon}^{\min}) \right]}_{\text{Bilateral Sharpness (BilS)}}. \end{aligned} \quad (10)$$

Equation (10) ensures that, during training, the loss not only decreases but also becomes more balanced on both sides of the current parameter point, resulting in a flatter parameter space around it.

1) *BilS Is a Better FI Than MaxS*: Fig. 1(a) illustrates the concepts of the MaxS, MinS, and BilS. MaxS only describes the flatness of the loss landscape on one side of the current parameter point. Due to the greedy nature of SAM, it seeks a flat region between the maximum loss point of the current optimization step and the current point. Consequently, the loss landscape may still be insufficiently flat after training, as illustrated in Fig. 1(b). In contrast, the proposed BilS is a new FI that identifies a flatter direction than SAM, making the parameter update direction more accurate and stable, as a comparison between Fig. 1(a) and (c).

Similar to (5), the gradient of MinS at the perturbation point $\mathbf{w} + \boldsymbol{\varepsilon}^{\min}$ can be written as follows:

$$\begin{aligned} \nabla_{\mathbf{w}} L^{\min}(\mathbf{w}) &\approx \nabla_{\mathbf{w}} L(\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}(\mathbf{w})) \\ &= \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}(\mathbf{w})} \\ &\quad + \frac{d\hat{\boldsymbol{\varepsilon}}^{\min}(\mathbf{w})}{d\mathbf{w}} \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}(\mathbf{w})}. \end{aligned} \quad (11)$$

To further accelerate computation, we also drop the second-order terms. The gradient of MinS is approximated as follows:

$$\nabla_{\mathbf{w}} L^{\min}(\mathbf{w}) \approx \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}}. \quad (12)$$

By combining the gradient of the training loss ($\nabla_{\mathbf{w}} L(\mathbf{w})$) with (6) and (12), the gradient for updating parameters in BSAM is as follows:

$$\begin{aligned} \nabla_{\mathbf{w}} \hat{L}(\mathbf{w}) &= \underbrace{\nabla_{\mathbf{w}} L(\mathbf{w})}_{\text{Gradient of the training loss}} \\ &+ \underbrace{(\nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\max}(\mathbf{w})} - \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}(\mathbf{w})})}_{\text{Gradients that promote flat minima}}. \end{aligned} \quad (13)$$

BSAM employs the base optimizer to perform gradient descent using the gradient from (13). The first term in this gradient promotes finding regions of local minima, while the second term encourages locating flatter regions.

2) *Solve GCP*: However, in the later stages of training, as illustrated in Fig. 2(a), a large fixed $\rho^{\min}$ can cause the perturbation point $\mathbf{w}^{\min}$ ($\mathbf{w}^{\min} = \mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}$) to overshoot the minimum point. The directions of $\nabla_{\mathbf{w}} L(\mathbf{w})$ and $\nabla_{\mathbf{w}} L(\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min})$ are opposite, causing the gradient conflict problem (GCP) between the gradient of the original point $\mathbf{w}$ and the gradient of the perturbed point $\mathbf{w}^{\min}$. Fig. 3 shows the cosine similarity between the gradients at $\mathbf{w}$ and $\mathbf{w} + \hat{\boldsymbol{\varepsilon}}^{\min}$ during ResNet-18 training with BSAM on CIFAR-100 dataset. In Fig. 3(a), with fixed $\rho^{\min}$, the cosine similarity is high in the early stages of training (e.g., before 60 000 iterations), indicating no gradient conflict. However, in the later stages of training (e.g., after 70 000 iterations), the cosine similarity decreases rapidly and may even drop below zero, indicating that the gradient conflict occurs.

Although gradient decomposition could address the GCP, we chose not to use it due to the additional computational overhead it introduces, which is undesirable in practical implementations. Instead, we observed that gradually decreasing $\rho^{\min}$ helps reduce gradient conflicts during training. We assume that the gradient within the local range of the parameter space at point $\mathbf{w}$ does not change abruptly. When $\rho^{\min}$ is small, the gradient at the perturbation point is close to that at $\mathbf{w}$, reducing the likelihood of gradient conflict. Fig. 2(b) illustrates that after

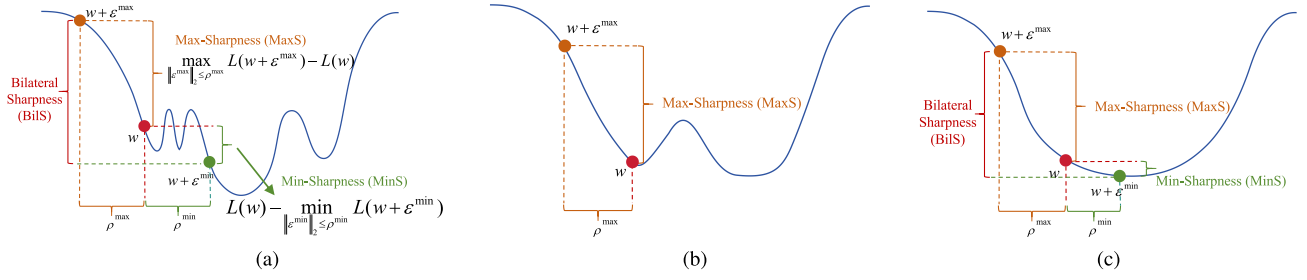


Fig. 1. Illustration of the notations of the MaxS, MinS, and BilS. (a) MaxS, MinS, and BilS during training. (b) Loss landscape after training with SAM. (c) Loss landscape after training with BSAM.

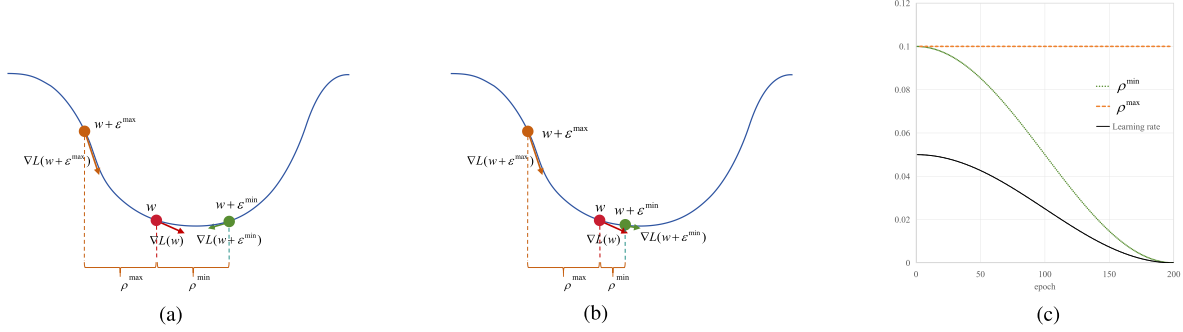


Fig. 2. Occurrence of gradient conflict under different $\rho^{\min}$ and the variation of $\rho^{\min}$ with learning rate in BSAM. (a) Gradient conflict occurs when $\rho^{\min}$ is too large. (b) There is no gradient conflict when $\rho^{\min}$ is small. (c) $\rho^{\min}$ changes with the learning rate.

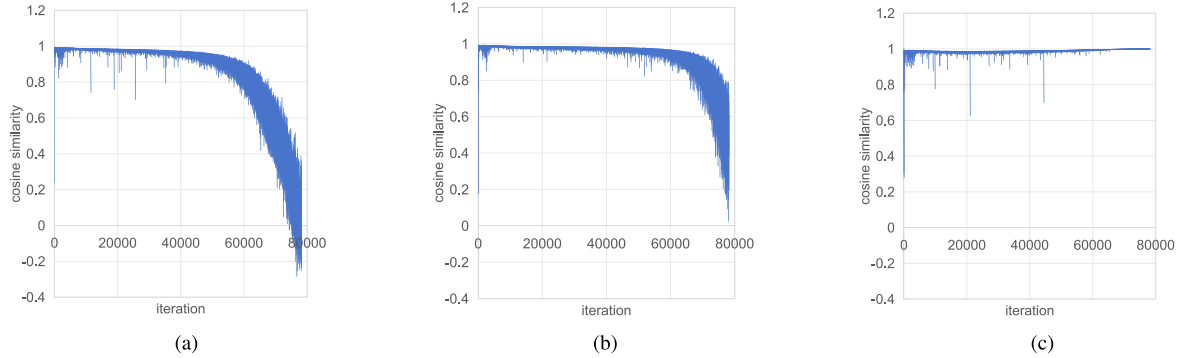


Fig. 3. Cosine similarity between the gradients at the point w and the point $w + \epsilon^{\min}$ in training stage under the different settings of $\rho^{\min}$. (a) Fixed $\rho^{\min}$. (b) $\rho^{\min}$ decreases from 0.1 to 0.05. (c) $\rho^{\min}$ decreases from 0.1 to 0.

decreasing $\rho^{\min}$ during training, there is no gradient conflict between $\nabla_w L(w)$ and $\nabla_w L(w + \epsilon^{\min})$. Based on this, we designed a scheduler that dynamically adjusts $\rho^{\min}$, decreasing it as the learning rate decreases. The scheduling process is defined as follows:

$$\rho_t^{\min} = \check{\rho}^{\min} + \frac{(\hat{\rho}^{\min} - \check{\rho}^{\min})(l_r - l_r^{\min})}{l_r^{\max} - l_r^{\min}} \quad (14)$$

where l_r is the learning rate at t th iteration, $l_r^{\max}$ and $l_r^{\min}$ are the maximum and minimum learning rates, respectively. $\rho_t^{\min}$ is the value of $\rho^{\min}$ at t th iteration, $\hat{\rho}^{\min}$ and $\check{\rho}^{\min}$ are hyper-parameters, representing the maximum and minimum value of $\rho_t^{\min}$, respectively. In our experimental setup described in Section IV-B1, the changes in $\rho_t^{\min}$ and learning rate during training are shown in Fig. 2(c). Comparing Fig. 3(b) and (c), the latter shows that the cosine similarity between gradients at w and $w^{\min}$ remains above 0, indicating no gradient conflict.

In fact, when the solution is near a minimum, the gradient $\nabla_w L(w)$ would theoretically be zero in (9). The magnitude of $\rho^{\min}$ has less impact on the perturbation than the size of the gradient. Therefore, reducing the $\rho_{\min}$ barely influences the results in (13).

3) *Balance Between the MaxS, MinS, and Training Loss:* For convenience, we denote the perturbation $\hat{\epsilon}$ in (4) as $\hat{\epsilon}^{\max}$. Our experiments reveal a disparity in the gradient magnitudes between the maximum perturbation parameter point $w^{\max}$ ($w^{\max} = w + \hat{\epsilon}^{\max}$) and the minimum perturbation parameter point $w^{\min}$ ($w^{\min} = w + \hat{\epsilon}^{\min}$) during training. Generally, the gradient magnitude at the maximum perturbation point is larger than that at the minimum point, which may diminish the effectiveness of MinS.

To address this, we follow the principle of maximum entropy. Concretely, we scale the gradient at the minimum point $\nabla_w L(w^{\min})$ to match the gradient magnitude at the maximum point $\nabla_w L(w^{\max})$, ensuring they work together to

Algorithm 1 Pseudocode of the Proposed Method

Require: The training dataset, the learning rate η , the batch size b , parameters ρ^{max} , $\hat{\rho}^{min}$ and $\check{\rho}^{min}$.

- 1: **for** $t = 1, 2, \dots$ **do**
- 2: Randomly sample a mini-batch $\mathbf{B}$ of size b ;
- 3: $\mathbf{g}_t = \nabla_{\mathbf{w}} L_{\mathbf{B}}(\mathbf{w}_t)$;
- 4: Compute perturbation $\hat{\mathbf{e}}_t^{max} = \rho_t^{max} \frac{\nabla_{\mathbf{w}} L(\mathbf{w}_t)}{\|\nabla_{\mathbf{w}} L(\mathbf{w}_t)\|}$;
- 5: Calculate the gradient at $\mathbf{w}_t + \hat{\mathbf{e}}_t^{max}$:
 $\mathbf{g}_t^{max} = \nabla_{\mathbf{w}} L_{\mathbf{B}}(\mathbf{w}_t + \hat{\mathbf{e}}_t^{max})$;
- 6: Compute perturbation $\hat{\mathbf{e}}_t^{min} = -\rho_t^{min} \frac{\nabla_{\mathbf{w}} L(\mathbf{w}_t)}{\|\nabla_{\mathbf{w}} L(\mathbf{w}_t)\|}$;
- 7: Calculate the gradient at $\mathbf{w}_t + \hat{\mathbf{e}}_t^{min}$:
 $\mathbf{g}_t^{min} = \nabla_{\mathbf{w}} L_{\mathbf{B}}(\mathbf{w}_t + \hat{\mathbf{e}}_t^{min})$;
- 8: Update the weights by:
 $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \left(\mathbf{g}_t + \mathbf{g}_t^{max} - \frac{\|\mathbf{g}_t^{max}\|}{\|\mathbf{g}_t^{min}\|} \mathbf{g}_t^{min} \right)$;
- 9: **end for**

promote generalization. In addition, we maintain the original scale between the gradient of the training loss and the gradient to encourage flatter minima. Finally, the gradient optimized at each step of BSAM can be rewritten as follows:

$$\nabla_{\mathbf{w}} \hat{L}(\mathbf{w}) = \nabla_{\mathbf{w}} L(\mathbf{w}) + \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{max}(\mathbf{w})} - \frac{\|\nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{max}(\mathbf{w})}\|}{\|\nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{min}(\mathbf{w})}\|} \cdot \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{min}(\mathbf{w})}. \quad (15)$$

In summary, Algorithm 1 outlines the proposed method.

4) *Time Complexity:* When the model has N weights and the batch size of optimizer is b , the time complexity of SGD is $O(bN)$. Assuming the majority of the time spent by the optimizer is on the forward and backward passes, SAM requires two forward and backward passes per optimization step, giving a complexity of $O(2bN)$. BSAM requires three forward and backward passes per optimization step, resulting in $O(3bN)$, making it approximately 1.5 times more computationally expensive than SAM. This increase in complexity is independent of dataset size, as it only impacts the forward and backward propagation steps, without requiring data reloading or additional data-dependent operations. However, in some scenarios, sacrificing efficiency is necessary to achieve better performance. Addressing this tradeoff remains an important direction for future research.

C. Example of the Training Characteristics of BSAM

Fig. 4 illustrates the loss and accuracy curves for both SAM and BSAM on the CIFAR-100 dataset. Two key observations can be made as follows.

- 1) The loss of BSAM decreases faster than that of SAM, while the accuracy of BSAM improves more rapidly than that of SAM. This indicates that the MinS guides the parameter update direction more accurately at the early stages of training. In addition, optimizing the MaxS and MinS with nonconflicting gradients accelerates the gradient descent process.
- 2) The loss curve of BSAM shows less fluctuation compared to SAM from epoch 0 to around 150. This behavior is explained by the fact that BSAM's trajectory follows a flatter direction than SAM's. It indicates that BSAM searches for flatter regions than SAM during the optimization process.

TABLE I

TEST ACCURACY (%) WITH DIFFERENT ρ^{min} ON CIFAR-10 AND CIFAR-100

ρ^{min}	CIFAR-10	CIFAR-100
0.01	96.75±0.09	81.56±0.05
0.025	96.78±0.02	81.29±0.11
0.05	96.82±0.12	81.65±0.04
0.1	96.66±0.13	81.48±0.18
0.25	96.18±0.09	81.58±0.25

D. Convergence Analysis

In this section, we study the convergence of BSAM. Since BSAM optimizes the training loss, MaxS, and MinS simultaneously, and employs gradient scaling during training, it is challenging to analyze the convergence of BSAM. The following assumptions on smoothness and bounded variance of stochastic gradients are standard in the literature on nonconvex optimization [10], [14].

Assumption 1 (Smoothness): $L(\mathbf{w})$ is τ -Lipschitz smooth in $\mathbf{w}$, i.e., $\|\nabla L(\mathbf{w}) - \nabla L(\mathbf{v})\| \leq \tau \|\mathbf{w} - \mathbf{v}\|$.

Assumption 2 (Bounded Variance of Stochastic Gradients): Given the training set $\mathbf{D}$, there exists $\mathbb{E}[\|\nabla L_{\mathbf{B}}(\mathbf{w}) - \nabla L_{\mathbf{D}}(\mathbf{w})\|^2] \leq \sigma^2$ for any data batch $\mathbf{B}$.

Theorem 1: Suppose that the true gradient at $\mathbf{w}$ is $\mathbf{g}_t = \nabla L(\mathbf{w}_t; \mathbf{D})$. Assumptions 1 and 2 hold. Let $\varsigma = \frac{(\|\nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{max}(\mathbf{w})}\|)/(\|\nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}^{min}(\mathbf{w})}\|)}{4(\rho^{max} + \rho^{min})^2 \tau^2} \leq 4(\rho^{max} + \rho^{min})^2 \tau^2$, then with learning rate $\eta_t = |(1 - \varsigma)/(\tau(1 + \varsigma^2))|/(\sqrt{T})$, we have the following bound for BSAM:

$$\sum_{t=0}^{T-1} \mathbb{E} \|\mathbf{g}_t\|^2 \leq \frac{L(\mathbf{w}_0) - \mathbb{E}[L(\mathbf{w}_{T+1})]}{Z_1 \sqrt{T}} + \frac{Z_1 Z_2}{\sqrt{T}} \quad (16)$$

where $Z_1 = |(1 - \varsigma)/(\tau(1 + \varsigma^2))|$, $Z_2 = (\tau\sigma^2 + \rho^{max^2} \tau^3 + \tau\varsigma^2 \sigma^2 + \tau\varsigma^2 \rho^{min^2} \tau^2)$. Z_1 and Z_2 are only depend on τ , ρ^{max} , ρ^{min} , σ .

Remark: Theorem 1 reveals that when T increases, the learning rate η and the gradient $\mathbf{g}_t$ gradually decreases, leading to the convergence of BSAM. In addition, when $\varsigma > 1$, a larger value of ς corresponds to a larger learning rate, which provides a larger gradient to accelerate convergence. However, excessively large learning rates in the later stages of training would hinder convergence. When $\varsigma < 1$, the smaller ς is, the larger Z_1 is, and the more conducive it is to convergence in the later stages of training. Therefore, our strategy of gradually reducing ρ^{min} promotes better convergence.

IV. EXPERIMENTAL RESULTS

A. Parameter Studies

The value of ρ^{min} affects whether a parameter point remains a minimum within a local region of $\mathbf{w}$ after perturbation of the gradient descent direction, which impacts the gradient direction and model performance. We study the effect of ρ^{min} on accuracy using Resnet-18 on the CIFAR-10 and CIFAR-100 datasets. The results are summarized in Table I. For CIFAR-10, when ρ^{min} is very small, the accuracy is close to that of SAM (as shown in Table II), because the gradients at the current parameter point and the perturbed point in the gradient descent direction are nearly identical, minimizing the

TABLE II
TEST ACCURACY (%) COMPARISON OF VARIOUS NETWORKS
ON CIFAR-10 AND CIFAR-100

ResNet-18	CIFAR-10	CIFAR-100
SGD	96.18±0.09	79.89±0.38
SAM	96.74±0.05	81.08±0.27
ASAM	96.63±0.15	81.68±0.12
FisherSAM	96.72±0.03	80.99±0.13
F-SAM	96.75±0.09	81.29±0.12
SGD-SALR	/	81.73±0.10
Unified VaSSO	96.34±0.01	80.01±0.07
SSAM-F	96.81±0.02	81.24±0.21
SSAM-D	96.87±0.04	80.58±0.44
LookSAM	96.47±0.13	80.48±0.24
ESAM	96.56±0.08	80.41±0.10
SAF	96.37±0.02	80.06±0.05
BSAM	96.82±0.12	81.48±0.18

WideResNet-28-10	CIFAR-10	CIFAR-100
SGD	96.93±0.05	82.56±0.27
SAM	97.52±0.05	84.71±0.21
ASAM	97.63±0.13	84.99±0.22
FisherSAM	97.46±0.18	84.91±0.07
F-SAM	97.53±0.11	85.16±0.07
SGD-SALR	/	83.12±0.09
Unified VaSSO	97.06±0.09	83.66±0.19
SSAM-F	97.71±0.23	85.16±0.96
SSAM-D	97.72±0.24	84.99±0.79
LookSAM	97.13±0.04	83.52±0.09
ESAM	97.29±0.11	84.51±0.01
SAF	97.08±0.15	83.81±0.04
BSAM	97.56±0.07	85.51±0.14

PyramidNet-110	CIFAR-10	CIFAR-100
SGD	97.10±0.08	83.38±0.21
SAM	97.65±0.06	86.06±0.16
ASAM ₍₃₀₀₎	97.82±0.07	86.47±0.09
FisherSAM ₍₃₀₀₎	97.64±0.09	86.53±0.07
F-SAM ₍₃₀₀₎	97.84±0.05	86.70±0.14
SGD-SALR	/	85.69±0.10
LookSAM	97.22±0.05	83.76±0.45
ESAM	97.81±0.01	85.56±0.05
SAF	97.34±0.06	84.71±0.01
BSAM	97.96±0.10	86.20±0.06

impact of MinS. When $\rho^{\min}$ is large (e.g., $\rho^{\min} = 0.1$ or 0.25), gradient conflicts as described in Section III-B may occur, reducing accuracy and performing worse than SAM. The best performance is achieved with $\rho^{\min} = 0.05$. For CIFAR-100, as the data complexity increases, using the values of ρ listed in the table for training the model consistently yields better accuracy than SAM. In summary, comparisons show that $\rho^{\min}$ is less sensitive in more complex scenarios.

B. Image Classification

1) *Setup*: To evaluate BSAM's effectiveness, we conduct experiments on the CIFAR-10 and CIFAR-100 [20] datasets across a variety of architectures, i.e., ResNet-18 [21], WideResNet-28-10 [22], and PyramidNet-110 [23]. These three models are trained for 200 epochs. For ResNet-18 and WideResNet-28-10, the initial learning rate is set to 0.05 with a cosine schedule, and the momentum and weight decay are set to 0.9 and 0.001. For PyramidNet-110, the initial learning rate is 0.1, the momentum and weight decay are set to 0.9 and 0.0005. For CIFAR-10, $\rho^{\max}$ and $\hat{\rho}^{\min}$ are set to 0.05, and for CIFAR-100, they are set to 0.1. $\check{\rho}^{\min}$ is set to 0.

We use vanilla SGD and SAM [1] as baselines. To comprehensively evaluate the performance, we have also included ASAM [8], FisherSAM [24], F-SAM [14], SGD-SALR [25], Unified VaSSO [26], SSAM-F, and SSAM-D [27] for comparison. These methods are the follow-up works of SAM that aim to enhance generalization. For ASAM, FisherSAM, and F-SAM, we report the results in [14]. It is worth noting that these three methods were run for 300 epochs for PyramidNet-110, which may lead to better results; we denote them as ASAM(300), FisherSAM(300), and F-SAM(300), respectively. For SGD-SALR and Unified VaSSO, we refer to the results presented in [25] and [26], respectively. For SSAM-F and SSAM-D, we refer to the results presented in [27]. In addition, we compared several ESAM variants, such as LookSAM [16], ESAM [9], and SAF [28]. We implemented LookSAM ourselves, while the results for ESAM and SAF are reported as presented in their original publications. For SGD, SAM, and BSAM, we conducted five independent runs with different seeds and reported the average accuracy and standard deviation.

To evaluate BSAM's performance on the large-scale benchmark, we conducted experiments on the ImageNet [29] dataset, which consists of 1000 classes, around 1.28 million training images, and 50 000 validation images, with an average resolution of 469×387 . Unlike CIFAR-10/100, ImageNet is better suited to reflect real-world scenarios and offers a more robust evaluation of a method's effectiveness. We resize the images to 224×224 for training ResNet-50 over 90 epochs, using the optimal hyperparameters for SGD, SAM, and BSAM as recommended by [9]. We conducted these experiments on ImageNet using an NVIDIA GeForce RTX 4090 GPU and compared the training speeds of BSAM and other methods.

2) *Results*: In Table II, BSAM outperforms both SGD and SAM on the test datasets. On the CIFAR-10 dataset, BSAM achieves the greatest improvement on PyramidNet-110, with increases of 0.86% and 0.31% over SGD and SAM, respectively. On the CIFAR-100 dataset, BSAM delivers the most significant improvement on WideResNet-28-10, with increases of 2.95% and 0.8% over SGD and SAM, respectively. This indicates that BSAM is suitable for various networks and datasets, but its ability to improve generalization performance varies across different networks and datasets.

In comparison to other SAM variants designed to enhance generalization, BSAM is more effective in most cases because it considers perturbations in the gradient descent direction. In addition, BSAM significantly outperforms ESAM methods, which often sacrifice model generalization performance for reduced computational costs. Although SSAM-D outperforms BSAM on CIFAR-10 (achieving 96.87%), it performs worse on CIFAR-100 with 80.58%, while BSAM achieves 81.48%. This indicates that BSAM is better at adapting to more complex datasets.

In general, BSAM achieved the best results on the test set in most cases on CIFAR-10 and CIFAR-100, indicating that BSAM enhances the model's generalization performance. This may be because BSAM considers both the gradient ascent and descent directions from the current parameter during training, which helps smooth the gradient descent process and ultimately guides the model toward flatter regions of the local minima.

TABLE III

TEST ACCURACY (%) AND TRAINING TIME (HOUR) OF RESNET-50 WITH SGD, SAM, AND BSAM ON THE IMAGENET

Methods	Accuracy	Training Time
SGD	76.00	63.5
SAM	76.70	100.2
SSAM-F	77.31	/
SSAM-D	77.25	/
BSAM	77.36	148.1

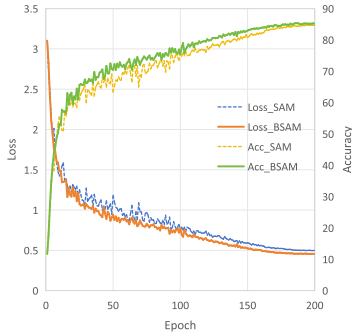


Fig. 4. Loss curves and accuracy ones of SAM and BSAM when WideResNet-28-10 was used.

Table III compares the test accuracy and training time of ResNet-50 trained with SGD, SAM, SSAM-F, SSAM-D, and BSAM on the ImageNet dataset. BSAM achieves the highest accuracy at 77.36%, outperforming both SGD and SAM. SSAM-F and SSAM-D also achieve competitive accuracies of 77.31% and 77.25%, but BSAM still leads, demonstrating that its optimization strategy is more effective in achieving better model performance. However, BSAM incurs a higher training time, approximately 1.5 times that of SAM (148.1 versus 100.2 h). This increase in training time aligns with the time complexity analysis in Section III-B. In scenarios where higher accuracy is required, the increased computational cost of BSAM is justified by its superior performance.

C. Top Eigenvalues of Hessian

The maximum Hessian eigenvalue is commonly viewed as a potential sharpness/flatness metric [5], [30], [31]. To offer a more quantitative assessment of sharpness compared to Fig. 5, we compare the maximum Hessian eigenvalues obtained from SGD, SAM, and BSAM. BSAM achieves smaller maximum eigenvalues on the test set, indicating that it finds flatter minima.

We analyze the Hessian spectra of ResNet-18 trained on CIFAR-100 with SGD, SAM, and BSAM using the CIFAR-100 test set. We employ Hutchinson's method [32] to compute the eigenvalues. For each method, we report the maximum eigenvalue of the Hessian, a histogram of the top-50 Hessian eigenvalues, and the mean and variance of these top-50 Hessian eigenvalues. As shown in Fig. 5, the distribution of the top-50 eigenvalues is significantly smaller for networks trained with SAM and BSAM compared to those trained with SGD. Specifically, the maximum eigenvalue for networks trained with BSAM is notably smaller than that for SGD, indicating that BSAM finds flatter minima and improves

TABLE IV

RESULTS OF RESNET-18 ON CIFAR-10 AND CIFAR-100 WITH DIFFERENT RATE OF LABEL NOISES

Datasets	Methods	Noise rate			
		20%	40%	60%	80%
CIFAR-10	SGD	85.94	70.25	48.50	28.91
	SAM	90.44	79.71	67.22	76.94
	BSAM	93.34	85.58	76.37	78.27
CIFAR-100	SGD	65.42	48.91	31.46	12.32
	SAM	69.10	55.06	35.59	9.82
	BSAM	71.65	57.61	38.71	32.69

generalization. BSAM also reduces the maximum eigenvalue compared to SAM. The mean and variance of the top-50 Hessian eigenvalues for the model trained with BSAM are both smaller than those for the models trained with SGD and SAM.

D. Robustness to Label Noise

Since prior studies have demonstrated that SAM is robust to label noise, this section evaluates the impact of applying BSAM in the classical noisy-label setting for CIFAR-10 and CIFAR-100. We evaluate the performance of BSAM under symmetric label noise by random flipping [33]. The training settings are the same as in Section IV-B1.

As shown in Table IV, for the CIFAR-10 dataset, we find that the accuracy of models optimized with SGD decreases rapidly with the increase in label noise rate, dropping to just 28.91% when the noise rate reaches 80%. In contrast, the accuracy of models optimized with SAM and BSAM decreases more slowly with increasing label noise, maintaining over 70% accuracy even when the noise rate reaches 80%. For CIFAR-100, we observe a similar trend, but when the noise rate reaches 80%, the accuracy of models trained with SAM is even lower than that of SGD. Fig. 6 shows the test accuracy of optimizing ResNet-18 on CIFAR-100 using SGD, SAM and BSAM when ρ is 0.05 and 0.1. When ρ_{max} is 0.1, SAM has difficulty converging stably when there are a large number of incorrect labels, as mentioned in [1]. However, the test accuracy of the models trained with BSAM steadily increases during training, indicating that BSAM can effectively converge even in the presence of a high noise rate. When we set ρ_{max} to 0.05, SAM is also able to converge. However, in the later stage of training, it is more prone to overfitting compared to BSAM. These observations suggest that BSAM ensures more stable convergence, facilitates reliable learning from clean labels, effectively mitigates the impact of noisy labels, and enhances generalization in high-noise environments. Overall, BSAM consistently enhances performance compared to SGD and SAM, confirming its superior generalization capability under label noise.

E. Transfer Learning

Transfer learning is a technique that leverages a model trained on one task to improve performance on a related task. By fine-tuning a pretrained model or extracting its features, it can significantly reduce the training time and data requirements for the new task, enhancing model performance even with limited data. Previous studies [1], [14] have demonstrated

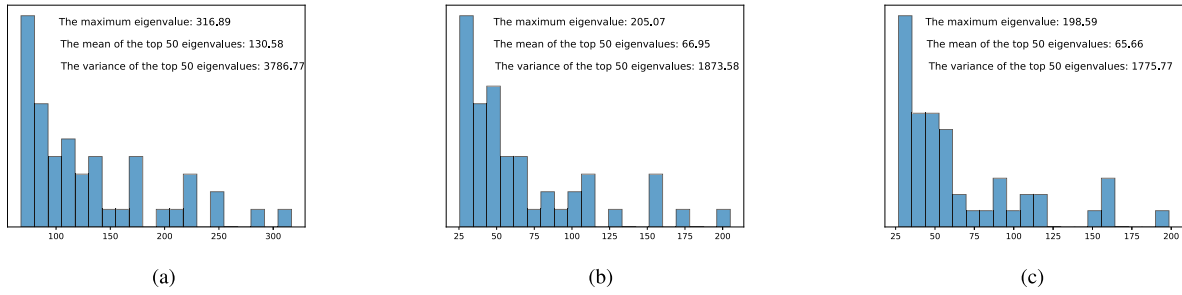


Fig. 5. Distribution of top-50 eigenvalues of hessian on the test set of CIFAR-100 with (a) SGD, (b) SAM, and (c) BSAM.

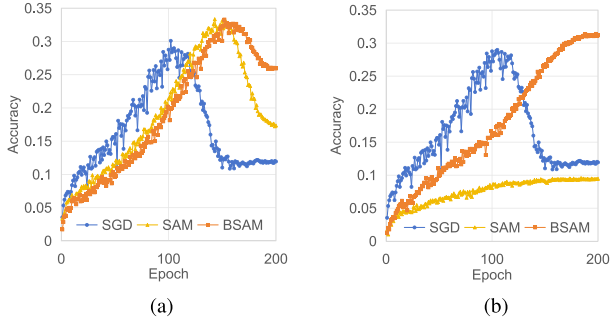


Fig. 6. Test accuracy of optimizing ResNet-18 on CIFAR-100 using SGD, SAM and BSAM when ρ_{max} is 0.05 and 0.1. (a) $\rho = 0.05$. (b) $\rho = 0.1$.

TABLE V
TEST ACCURACY (%) RESULTS OF FINE-TUNING
ON DIFFERENT DATASETS

Efficientnet_b0	SGD	SAM	BSAM
CIFAR-10	97.32 $\pm$ 0.07	97.48 $\pm$ 0.07	97.52$\pm$0.04
CIFAR-100	87.05 $\pm$ 0.11	87.27 $\pm$ 0.21	87.40$\pm$0.21
Flowers102	77.11 $\pm$ 0.36	78.04 $\pm$ 0.37	79.02$\pm$1.10
Stanford Cars	74.47 $\pm$ 0.55	75.75 $\pm$ 0.41	77.54$\pm$0.62
OxfordIIITPet	87.89 $\pm$ 0.43	88.30 $\pm$ 0.16	89.42$\pm$0.25
ResNet-50	SGD	SAM	BSAM
CIFAR-10	97.28 $\pm$ 0.05	97.78 $\pm$ 0.02	97.81$\pm$0.06
CIFAR-100	86.20 $\pm$ 0.18	87.48 $\pm$ 0.04	87.51$\pm$0.05
Flowers102	86.74 $\pm$ 0.16	87.53 $\pm$ 0.13	88.22$\pm$0.20
Stanford Cars	80.02 $\pm$ 0.24	83.69 $\pm$ 0.19	84.36$\pm$0.16
OxfordIIITPet	93.11 $\pm$ 0.17	93.14$\pm$0.14	93.06 $\pm$ 0.15

the effectiveness of SAM and its variants in transfer learning. In this section, we evaluate the performance of BSAM in transfer learning tasks.

Specifically, we apply SGD, SAM, and BSAM to fine-tune EfficientNet-b0 and ResNet-50 (pretrained on ImageNet). Weights are initialized using the values from publicly available checkpoints, except for the final classification layer, which is resized to accommodate the new number of classes and is initialized randomly. We trained two models for 30 epochs with a batch size of 128. The initial learning rate is set to 0.01 with cosine learning rate decay. Weight decay is set to $1e^{-5}$ for EfficientNet-b0 and $1e^{-4}$ for ResNet-50 as mentioned in their papers. For SAM and BSAM, we use SGD as the base optimizer; ρ , ρ^{max} , and $\hat{\rho}^{min}$ are all set to 0.05. We do not use any data augmentations for Flowers102 [34], Stanford Cars [35], OxfordIIITPet [36]. For CIFAR-10/100, we employ the same data augmentations as previous experiments.

As seen in Table V, SAM exhibits stronger generalization capabilities in transfer learning compared to SGD on the

EfficientNet-b0 and ResNet-50, and BSAM further improves SAM. BSAM achieves optimal results in transfer learning tasks across multiple datasets for both models. Notably, it shows significant improvements on smaller datasets such as Flowers102 and Stanford Cars.

F. Application to Human Pose Estimation

2-D human pose estimation (2-D HPE) is a computer vision task that involves detecting and analyzing human poses from images or videos. We apply SAM and BSAM to 2-D HPE to evaluate their general applicability, using the SimCC method [37] for validation. The key idea of SimCC is to treat human pose estimation as two classification tasks, one for vertical and one for horizontal coordinates. We use SGD, SAM, and BSAM as optimizers for the SimCC, and conduct experiments on the MPII dataset [38]. The head-normalized probability of correct keypoint (PCKh) [38] score is used for model evaluation. All experiments are conducted with the input image size 256×256 , and the batch size is set to 64.

The results are shown in Table VI, revealing that BSAM performs better on key body joints such as elbow, wrist, and hip, while showing a decline in performance on other joints. Overall, BSAM achieves better average results compared to SGD and SAM. It's worth noting that both SAM and BSAM use the Adam optimizer as their base optimizer in these experiments. This indicates that BSAM remains effective even when Adam is used as the base optimizer.

G. Application to Semantic Segmentation

We conduct semantic segmentation experiments on the PASCAL VOC 2012 dataset [39], which includes 20 object categories and one background class. Using PSPNet [40] and DeepLabv3+ [41] as representative models, we follow [40]'s procedure with augmented data from [42], resulting in 10 582 training, 1449 validation, and 1456 test images. We compare our proposed BSAM with SGD and SAM.

Table VII presents the results of semantic segmentation on the VOC 2012 dataset using different optimization methods. The results demonstrate BSAM's consistent superiority across both architectures. It shows particular strength on DeepLabv3+, surpassing SAM by 0.99%. These consistent improvements demonstrate BSAM's superior generalization across different architectures.

H. Application to QAT

We employ BSAM as the optimizer for quantization-aware training (QAT) [43] to demonstrate BSAM's broader appli-

TABLE VI
RESULTS OF TRAINING SIMCC ON MPII WITH ADAM, SAM, AND BSAM

Methods	Head	Shoulder	Elbow	Wrist	Hip	Knee	Ankle	Mean
SimCC+Adam	96.828	95.771	89.654	84.272	88.679	85.151	81.436	89.318
SimCC+SAM	96.965	95.703	89.637	84.222	88.402	85.513	81.862	89.388
SimCC+BSAM	96.794	95.584	89.995	85.317	89.008	85.392	81.082	89.552

TABLE VII
MEAN IOU (%) OF SEMANTIC SEGMENTATION WITH SGD, SAM, AND BSAM ON THE VOC 2012

Methods	PSPNet	DeepLabv3+
SGD	78.70±0.28	77.12±0.28
SAM	79.10±0.14	77.25±0.16
BSAM	79.20±0.14	77.96±0.39

TABLE VIII
TEST ACCURACY (%) RESULTS OF QAT WITH SGD, SAM, AND BSAM ON THE CIFAR-10

Methods	ResNet-18	MobileNetV1
Full prec.	88.72	85.81
SGD	88.86±0.18	84.04±0.13
SAM	89.75±0.21	84.72±0.11
BSAM	89.84±0.11	84.76±0.10

capability. We applied SGD, SAM, and BSAM algorithms to quantize the parameters of the ResNet-18 and MobileNetV1 models to W4A4 on the CIFAR-10 dataset; the results are presented in Table VIII. We found that on ResNet-18, the quantized models trained with SAM and BSAM achieved results that were higher than those of the models trained with SGD, even surpassing the accuracy of the full-precision model. On MobileNetV1, SAM and BSAM also achieved better results than SGD. Overall, both SAM and BSAM enhance the generalization performance of quantized models, with BSAM achieving marginally better results than SAM.

V. CONCLUSION

In this article, we address the limitations of SAM, particularly its focus solely on sharpness along the gradient ascent direction. By identifying the FIP in SAM, we propose a novel approach that integrates MinS alongside MaxS to create a more effective BilS indicator. Then, we introduce bilateral BSAM. The bilateral optimization strategy, as detailed in (10), ensures the loss landscape is flattened on both sides of the current parameter point, leading to flatter minima. Experiments across various models and datasets show that BSAM outperforms SGD and SAM, indicating that BSAM can improve the model's generalization performance. In addition, we apply BSAM to a variety of tasks. Experiments on label noise demonstrate that BSAM exhibits superior robustness to label noise. Experiments on transfer learning, human pose estimation, semantic segmentation, and model quantization, BSAM achieved better results than both SGD and SAM, indicating that BSAM can be easily applied to a variety of tasks and has broad applicability.

In future work, we will further explore the impact of perturbation directions in SAM on the optimization process, with the goal of identifying flatter regions of the minimum to enhance generalization performance. In addition, we will also

explore strategies to improve the efficiency of BSAM, thereby increasing its practical applicability.

REFERENCES

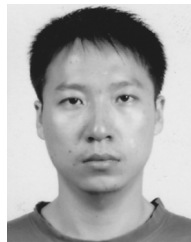
- [1] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, "Sharpness-aware minimization for efficiently improving generalization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 2315–2333.
- [2] H. Li, Z. Xu, G. Taylor, C. Studer, and T. Goldstein, "Visualizing the loss landscape of neural nets," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 6391–6401.
- [3] B. Neyshabur, S. Bhojanapalli, D. McAllester, and N. Srebro, "Exploring generalization in deep learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 5947–5956.
- [4] Y. Jiang, B. Neyshabur, H. Mobahi, D. Krishnan, and S. Bengio, "Fantastic generalization measures and where to find them," 2019, *arXiv:1912.02178*.
- [5] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, "On large-batch training for deep learning: Generalization gap and sharp minima," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 2874–2889.
- [6] F. He, T. Liu, and D. Tao, "Why ResNet works? Residuals generalize," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 12, pp. 5349–5362, Dec. 2020.
- [7] L. Wu et al., "How sgd selects the global minima in over-parameterized learning: A dynamical stability perspective," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 8279–8288.
- [8] J. Kwon, J. Kim, H. Park, and I. K. Choi, "ASAM: Adaptive sharpness-aware minimization for scale-invariant learning of deep neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5905–5914.
- [9] J. Du et al., "Efficient sharpness-aware minimization for improved training of neural networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 9703–9720.
- [10] J. Zhuang et al., "Surrogate gap minimization improves sharpness-aware training," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 2553–2576.
- [11] M. Dong, Y. Yang, K. Zeng, Q. Wang, and T. Shen, "Implicit sharpness-aware minimization for domain generalization," *Remote Sens.*, vol. 16, no. 16, p. 2877, Aug. 2024.
- [12] Y. Zhao, H. Zhang, and X. Hu, "Penalizing gradient norm for efficiently improving generalization in deep learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 26982–26992.
- [13] X. Zhang, R. Xu, H. Yu, H. Zou, and P. Cui, "Gradient norm aware minimization seeks first-order flatness and improves generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20247–20257.
- [14] T. Li, P. Zhou, Z. He, X. Cheng, and X. Huang, "Friendly sharpness-aware minimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5631–5640.
- [15] Y. Zhao, H. Zhang, and X. Hu, "Randomized sharpness-aware training for boosting computational efficiency in deep learning," 2022, *arXiv:2203.09962*.
- [16] Y. Liu, S. Mai, X. Chen, C.-J. Hsieh, and Y. You, "Towards efficient and scalable sharpness-aware minimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 12360–12370.
- [17] W. Jiang, H. Yang, Y. Zhang, and J. Kwok, "An adaptive policy to employ sharpness-aware minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023, pp. 21681–21698.
- [18] R. Ni, P.-Y. Chiang, J. Geiping, M. Goldblum, A. G. Wilson, and T. Goldstein, "K-SAM: Sharpness-aware minimization at the speed of SGD," 2022, *arXiv:2210.12864*.
- [19] M. Becker, F. Altroch, and B. Risse, "Momentum-SAM: Sharpness-aware minimization without computational overhead," 2024, *arXiv:2401.12033*.
- [20] A. Krizhevsky et al., "Learning multiple layers of features from tiny images," Citeseer, 2009, pp. 1–60.

- [21] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [22] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2016, *arXiv:1605.07146*.
- [23] D. Han, J. Kim, and J. Kim, "Deep pyramidal residual networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5927–5935.
- [24] M. Kim, D. Li, S. X. Hu, and T. Hospedales, "Fisher sam: Information geometry and sharpness aware minimisation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 11148–11161.
- [25] X. Yue, M. Nouiehed, and R. Al Kontar, "SALR: Sharpness-aware learning rate scheduler for improved generalization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 1–10, Sep. 2024, doi: [10.1109/TNNLS.2023.3263393](https://doi.org/10.1109/TNNLS.2023.3263393).
- [26] D. Oikonomou and N. Loizou, "Sharpness-aware minimization: General analysis and improved rates," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025, pp. 15149–15175.
- [27] P. Mi et al., "Systematic investigation of sparse perturbed sharpness-aware minimization optimizer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 10, pp. 8538–8549, Oct. 2025.
- [28] J. Du, D. Zhou, J. Feng, V. Tan, and J. T. Zhou, "Sharpness-aware training for free," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 23439–23451.
- [29] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
- [30] S. Jastrzebski et al., "Three factors influencing minima in SGD," 2017, *arXiv:1711.04623*.
- [31] P. Chaudhari et al., "Entropy-SGD: Biasing gradient descent into wide valleys*," *J. Stat. Mechanics: Theory Exp.*, vol. 2019, no. 12, Dec. 2019, Art. no. 124018.
- [32] Z. Yao, A. Gholami, K. Keutzer, and M. W. Mahoney, "PyHessian: Neural networks through the lens of the Hessian," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2020, pp. 581–590.
- [33] J. Huang, L. Qu, R. Jia, and B. Zhao, "O2U-Net: A simple noisy label detection approach for deep neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3326–3334.
- [34] M.-E. Nilsback and A. Zisserman, "Automated flower classification over a large number of classes," in *Proc. 6th Indian Conf. Comput. Vis., Graph. Image Process.*, Dec. 2008, pp. 722–729.
- [35] J. Krause, M. Stark, J. Deng, and L. Fei-Fei, "3D object representations for fine-grained categorization," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Dec. 2013, pp. 554–561.
- [36] O. M. Parkhi, A. Vedaldi, A. Zisserman, and C. V. Jawahar, "Cats and dogs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3498–3505.
- [37] Y. Li et al., "SimCC: A simple coordinate classification perspective for human pose estimation," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 89–106.
- [38] M. Andriluka, L. Pishchulin, P. Gehler, and S. Bernt, "2D human pose estimation: New benchmark and state of the art analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3686–3693.
- [39] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
- [40] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2881–2890.
- [41] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
- [42] B. Hariharan, P. Arbeláez, L. Bourdev, S. Maji, and J. Malik, "Semantic contours from inverse detectors," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 991–998.
- [43] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, 2018, pp. 2704–2713.
- [44] S. K. Esser, J. L. McKinstry, D. Bablani, R. Appuswamy, and D. S. Modha, "Learned step size quantization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020, pp. 6910–6921.
- [45] X. Wei, R. Gong, Y. Li, X. Liu, and F. Yu, "Qdrop: Randomly dropping quantization for extremely low-bit post-training quantization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 2831–2849.
- [46] M. Nagel, M. Fourmarakis, Y. Bondarenko, and T. Blankevoort, "Overcoming oscillations in quantization-aware training," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 16318–16330.



Jiaxin Deng received the B.S. degree in digital media technology from Three Gorges University, Yichang, China, in 2017, and the M.S. degree in computer science from Guizhou Normal University, Guiyang, China, in 2021. He is currently pursuing the Ph.D. degree in control engineering and science with Beijing University of Technology, Beijing, China.

His research interests include deep learning, computer vision, and model optimization.



Junbiao Pang received the B.S. and M.S. degrees in computational fluid dynamics and computer science from Harbin Institute of Technology, Harbin, China, in 2002 and 2004, respectively, and the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2011.

His research interests include multimedia and machine learning for transportation problem.



Baochang Zhang has long been engaged in the research of fast small target detection and recognition in complex backgrounds, and is currently paying special attention to the combination of control theory and deep learning. He has trained nearly ten excellent papers at the school and college levels.

Mr. Zhang is a member of the program committee of top conferences such as CVPR, ICCV, and *ACM Microwave Magazine*. He is an Editorial Board Member of *IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS* and *IJARS*.



Qingming Huang (Fellow, IEEE) received the B.S. degree in computer science and the Ph.D. degree in computer engineering from Harbin Institute of Technology, Harbin, China, in 1988 and 1994, respectively.

He is currently the Chair Professor with the School of Computer Science and Technology, University of Chinese Academy of Sciences, Beijing, China. He has published more than 600 academic papers in prestigious international journals including *IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE*, *IEEE TRANSACTIONS ON IMAGE PROCESSING*, and *IEEE TRANSACTIONS ON MULTIMEDIA*, and top-level conferences such as ICML, NeurIPS, ICCV, CVPR, and ACM Multimedia. His research interests include multimedia computing, image/video processing, machine learning, and computer vision.

Dr. Huang was an Associate Editor of *IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY* and *Acta Automatica Sinica*.